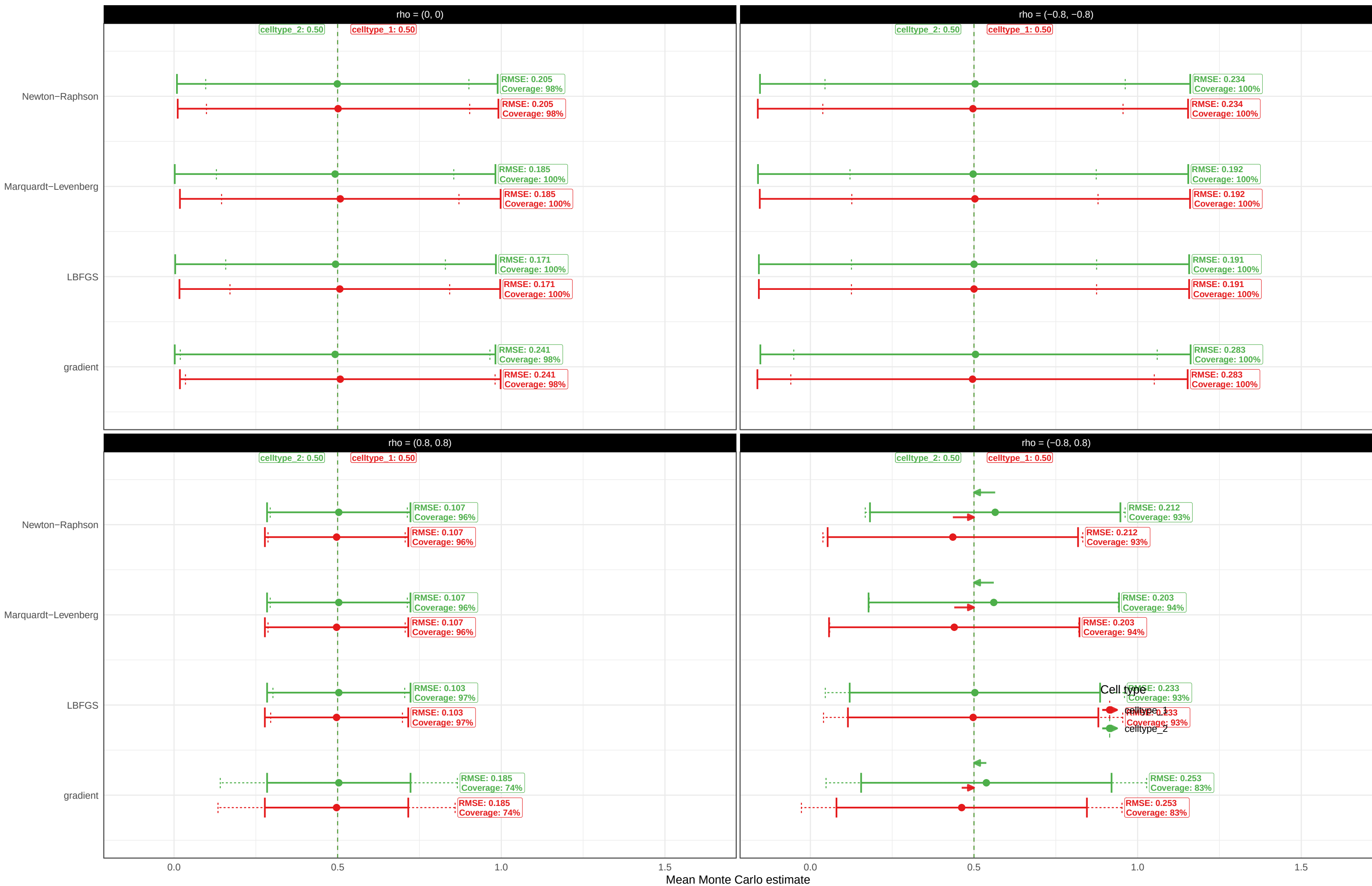
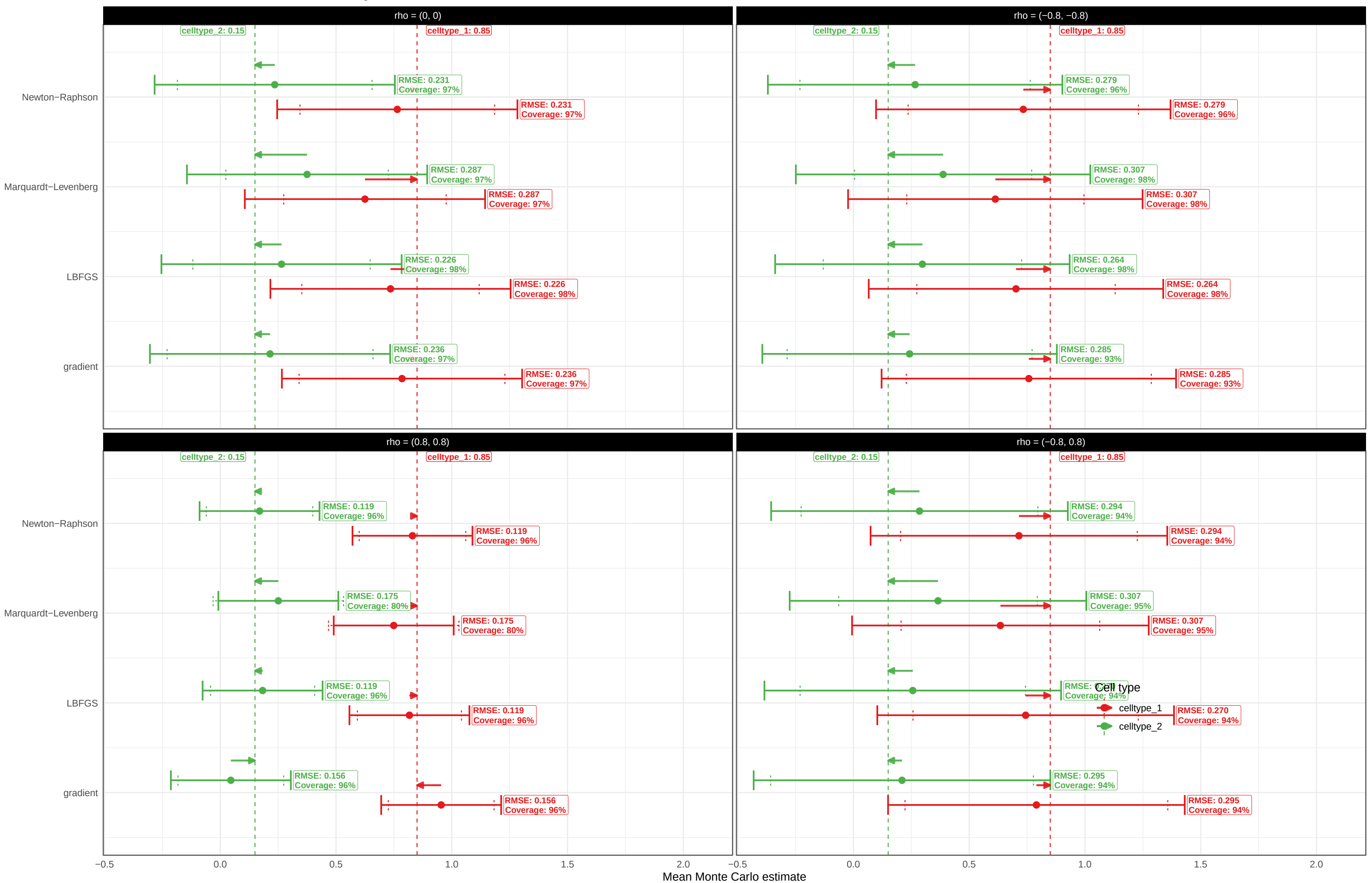
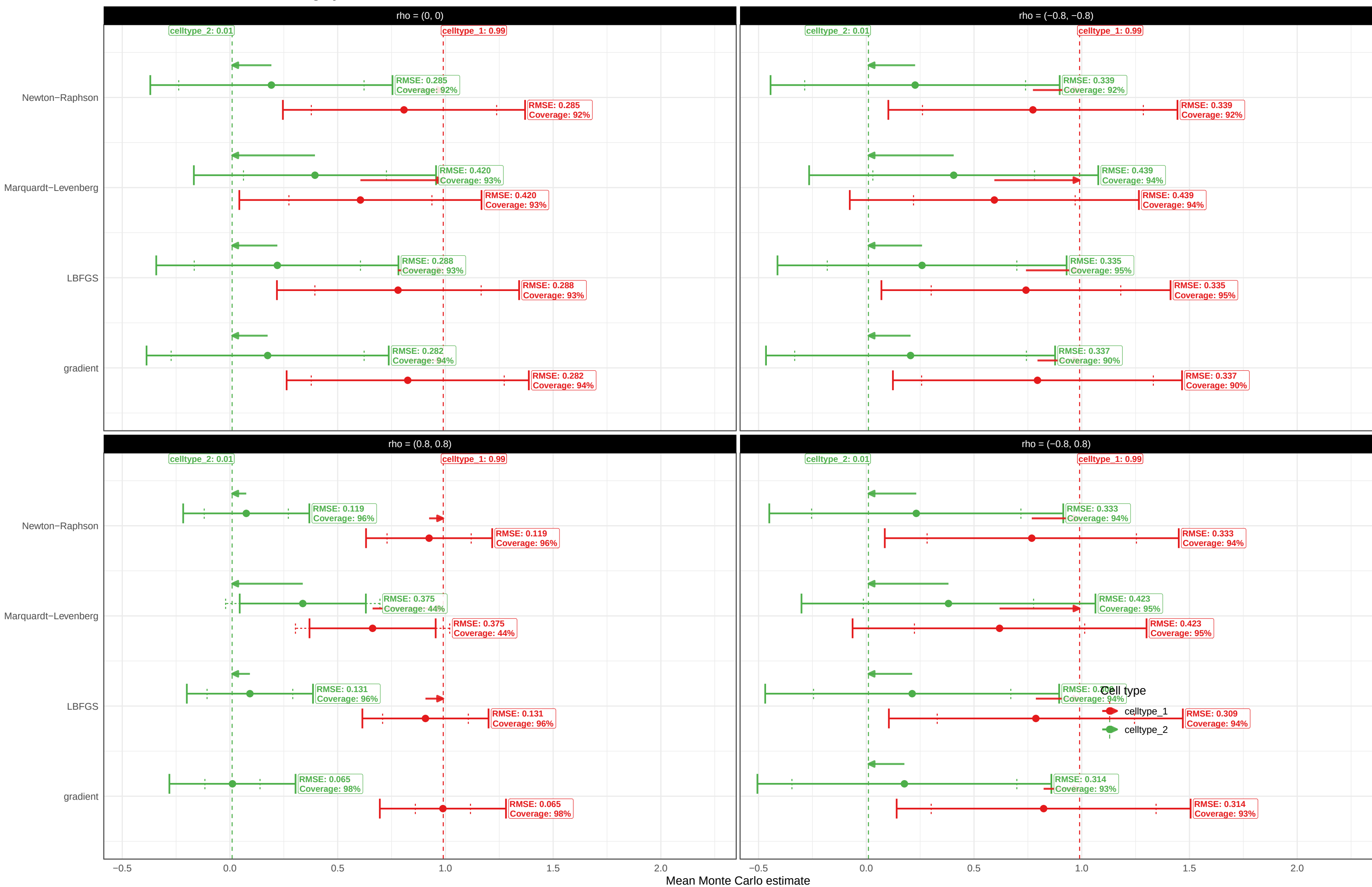




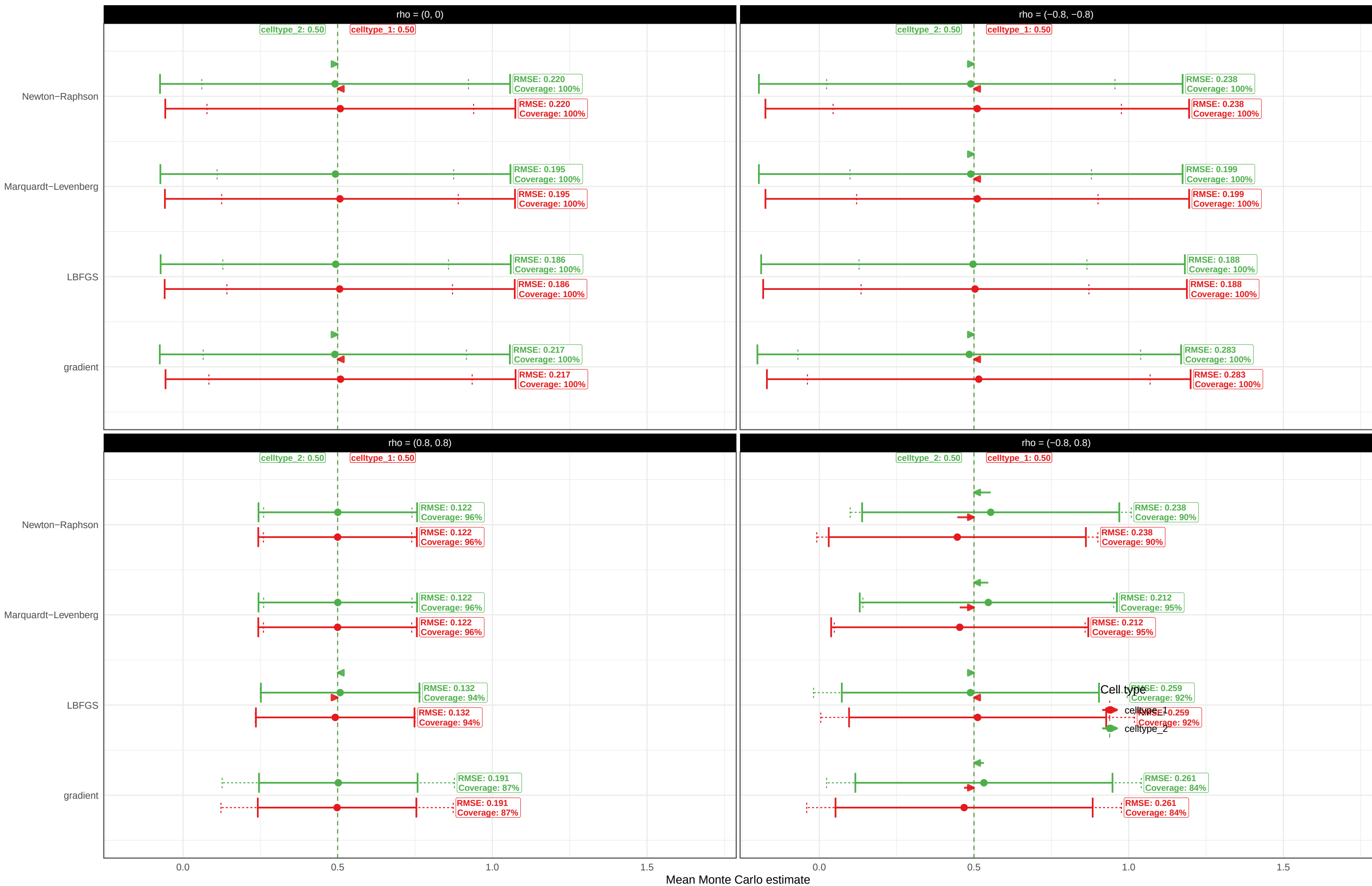
small_CLD / homoscedastic / moderately unbalanced



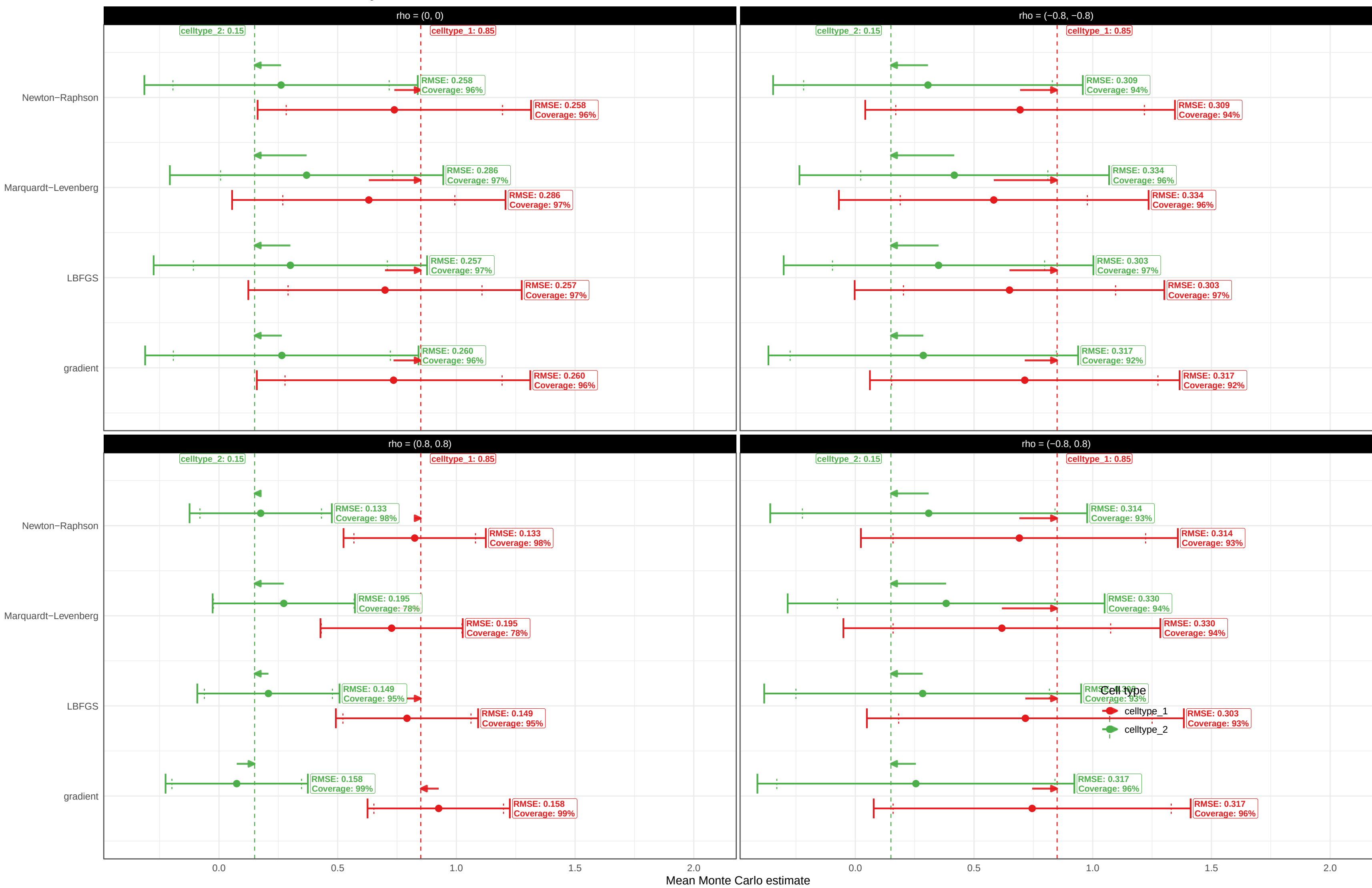
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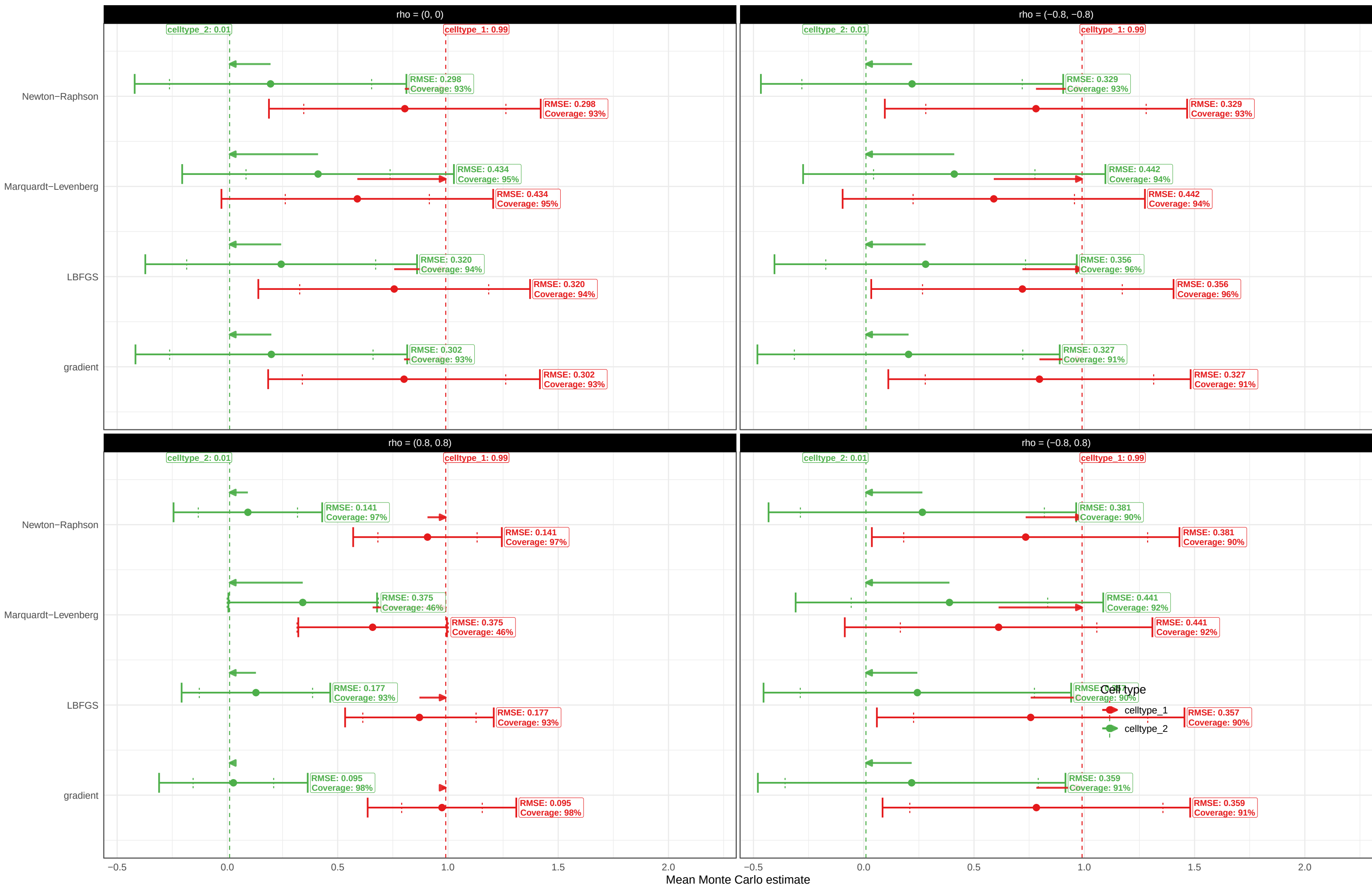


small_CLD / heteroscedastic / balanced

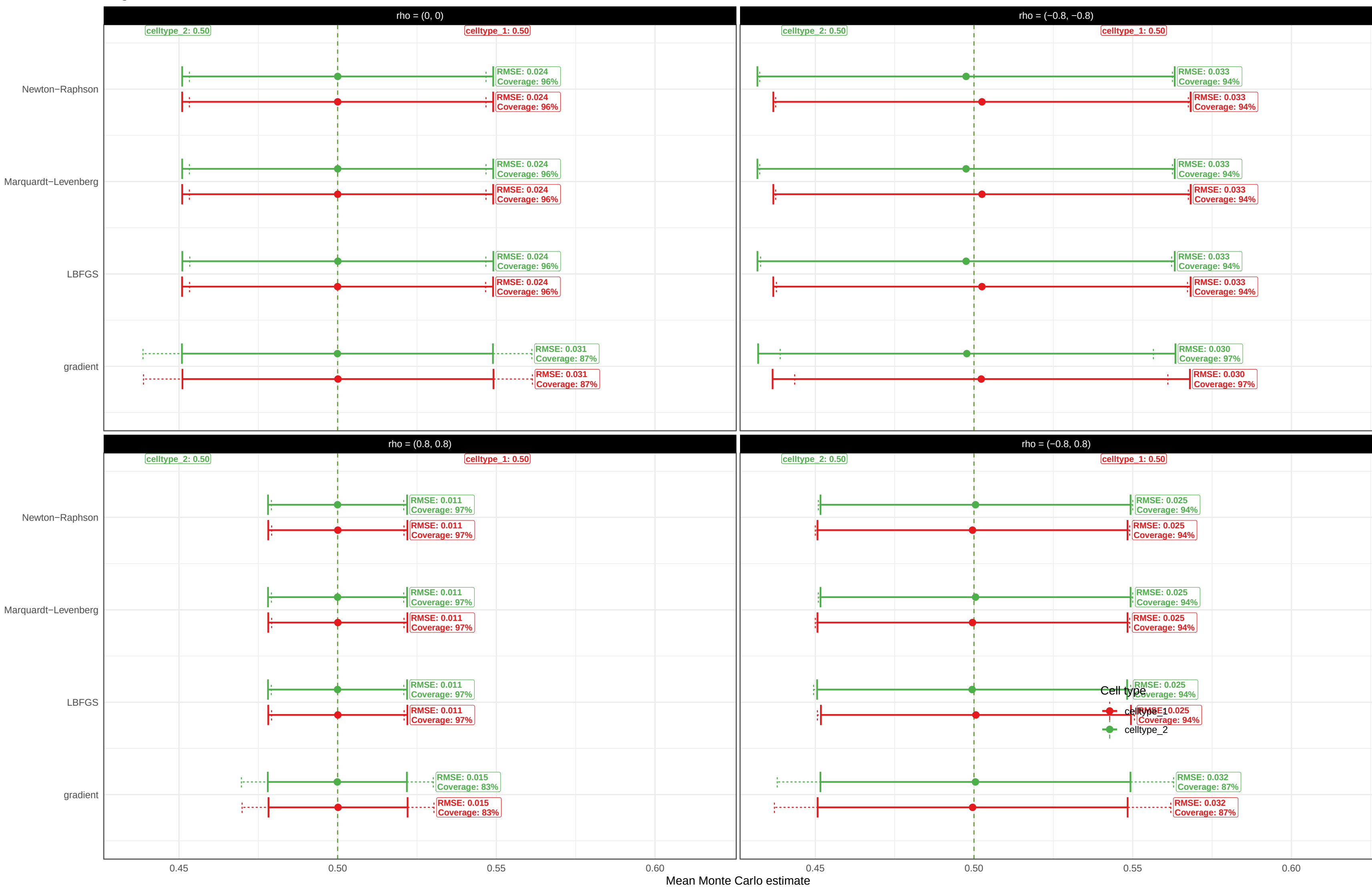


small_CLD / heteroscedastic / moderately unbalanced

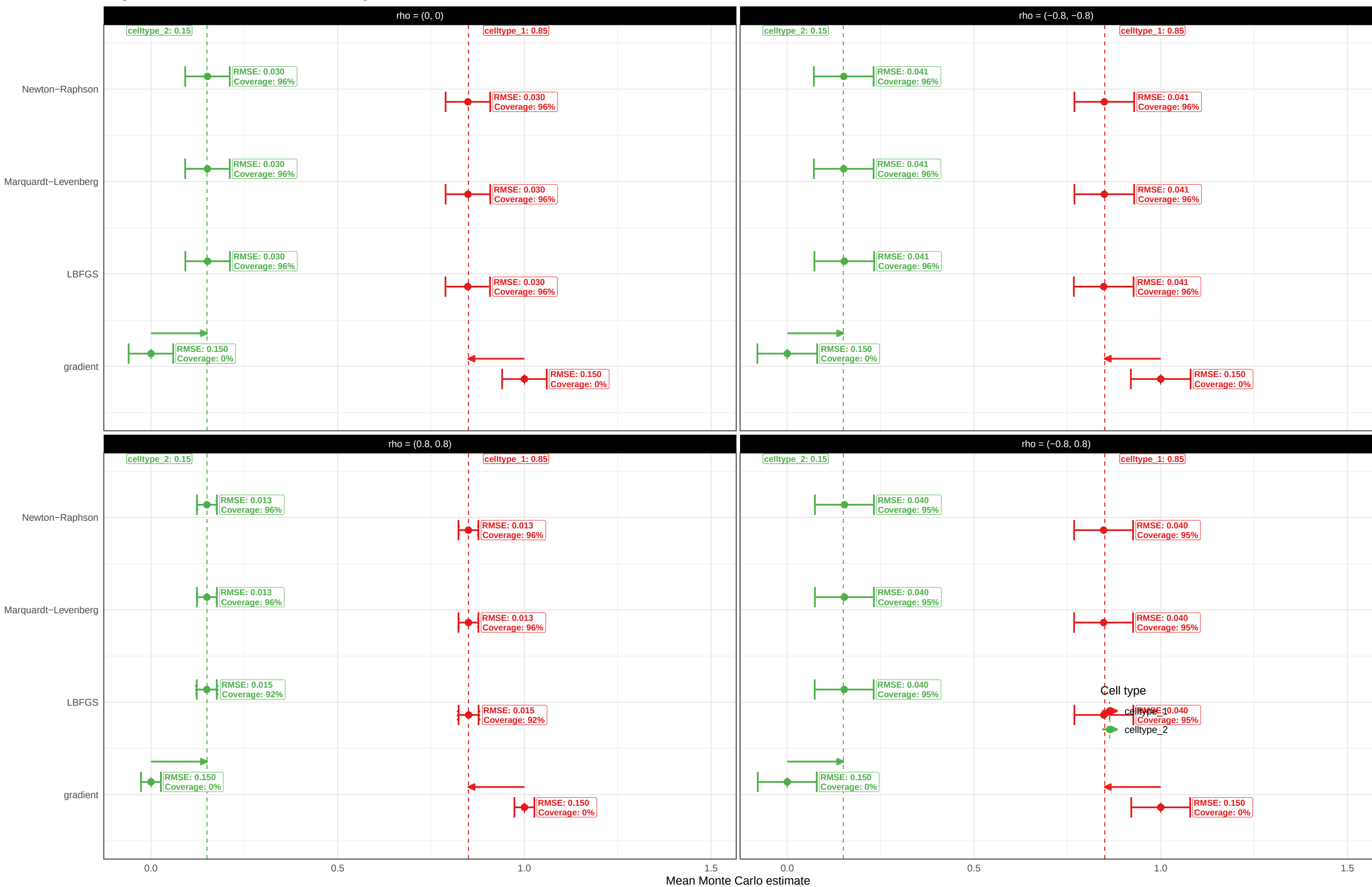




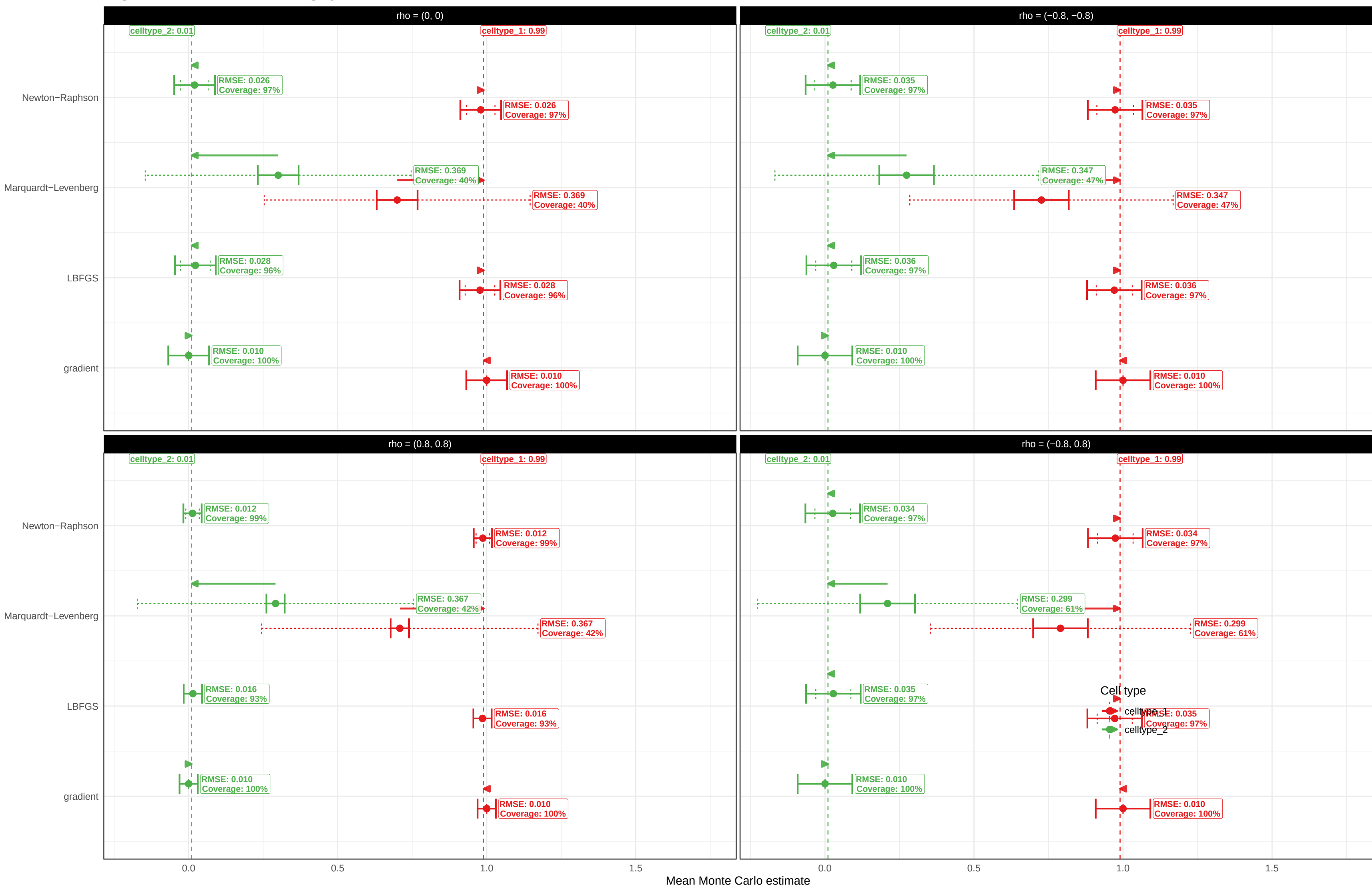
large_CLD / homoscedastic / balanced



large_CLD / homoscedastic / moderately unbalanced



large_CLD / homoscedastic / highly unbalanced



Solid whiskers: mean estimate plus or minus 1.96 times the expected-Fisher Wald SE at the true composition (confint.decovart_fit / vcov_ill_delta). Dashed whiskers: plus or minus 1.96 times the empirical Monte Carlo SD. Horizontal arrows: bias toward the true proportion. Top labels: true cell-type proportions (abundant type to the right of the other).

large_CLD / heteroscedastic / balanced

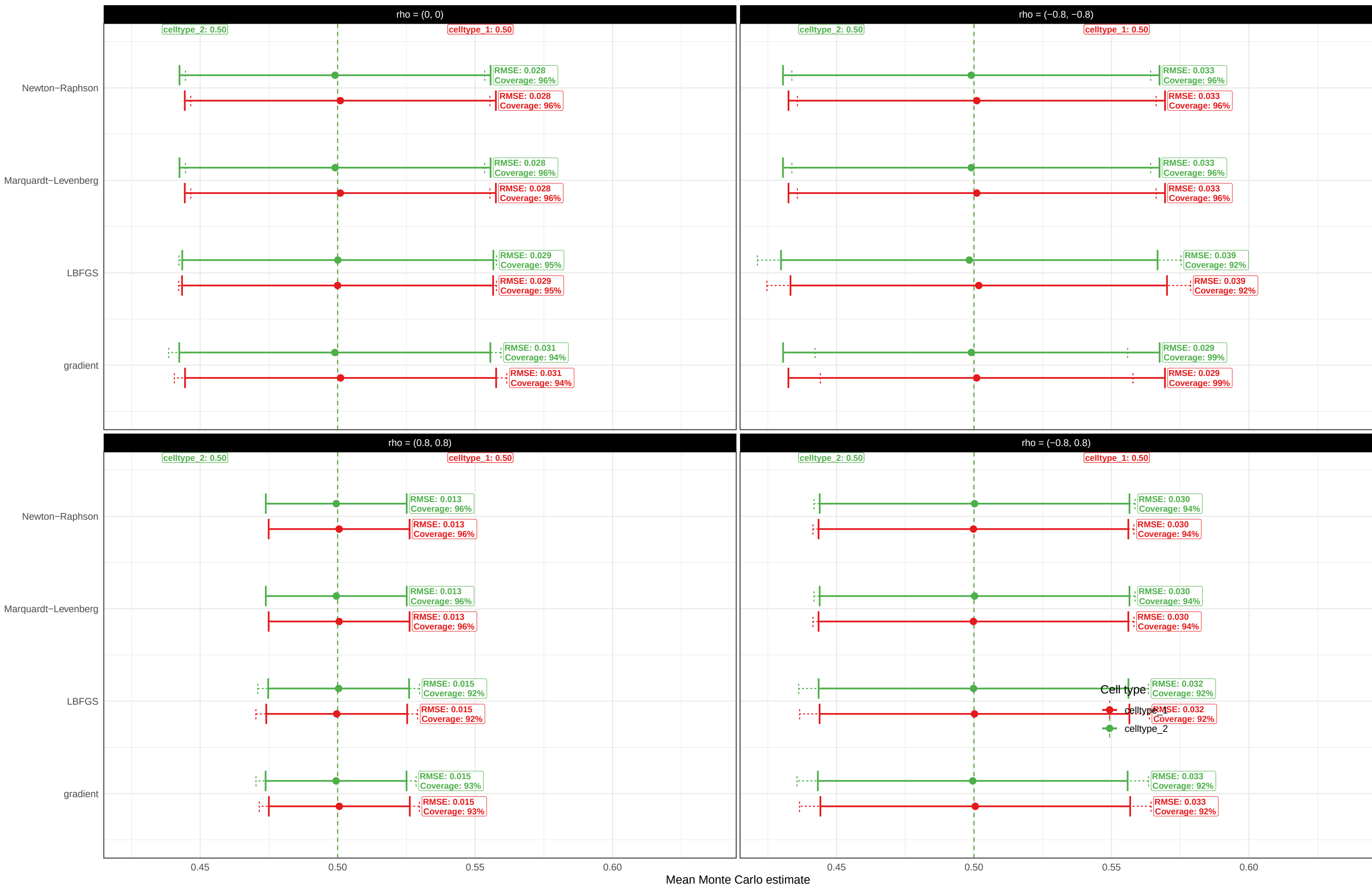


Figure 1 displays four panels showing the Root Mean Square Error (RMSE) and Coverage for celltype 1 and celltype 2 across different values of ρ . The panels are arranged in a 2x2 grid, corresponding to $\rho = (0, 0)$, $\rho = (-0.8, -0.8)$, $\rho = (0.8, 0.8)$, and $\rho = (-0.8, 0.8)$.

Each panel shows the RMSE and Coverage for celltype 1 (red) and celltype 2 (green). The RMSE values are displayed next to the data points, and the Coverage percentages are shown in the legend. The legend indicates that celltype 1 is represented by red and celltype 2 by green.

The RMSE values for celltype 1 are consistently 0.035, 0.041, 0.047, and 0.150 across the panels. The Coverage percentages for celltype 1 are 95%, 96%, 94%, and 0% respectively. The RMSE values for celltype 2 are consistently 0.035, 0.041, 0.047, and 0.150 across the panels. The Coverage percentages for celltype 2 are 95%, 96%, 94%, and 0% respectively.

Solid whiskers: mean estimate plus or minus 1.96 times the expected-Fisher Wald SE at the true composition ($\text{confint.decovart_fit} / \text{vcov_ilr_delta}$). Dashed whiskers: plus or minus 1.96 times the empirical Monte Carlo SD. Horizontal arrows: bias toward the true proportion. Top labels: true cell-type proportions (abundant type to the right).

large_CLD / heteroscedastic / highly unbalanced

